Xiyao Long

Disciplined Inconsistency with Consistency Types

The authors start to introduce the motivation of their paper by discussing in real-world applications, implementation variating consensus for different operations are important for making an efficient program in handling data while ensure safety/ correctness of the program. For example, for an online ticketing system, it's important to enable strongly consistent storage for financial transaction part, yet, tolerating certain error in reading access in how many seats left for a show can be acceptable and would make a more efficient program.

However, this mixed handling need to be implemented correctly else it can yield problems that greatly compromise the use of program, such as if we want to enable the ticketing system to charge a higher price for the last n seats, using an not-up-to-date value can cause the revenue to be below the expected.

They introduce the IPA programming model which adopt Abstract data type (ADT). The members in the ADT can be operations and they can be annotated with different consistency levels (strong, weak etc). The consistency types or policy can be applied to ADT instance for its lifetime and they can be either static or dynamic. A main motivation of this model is that by introducing such annotation, it enables enforcement to prevent the inconsistent values form flowing into the consistent values. The specific types returned are generally dependent on the nature of the operation, strong consistency reads return the strongest consistency type in the primitive T while weak reads return weak consistency type. They further discussed how they enabled operation that bound the error of an operation using the IPA programming model. I think the IPA paper has introduced the implementation side of the distributed system with application and is very informative.